

Laboratory Report

Course ID: CPS 3498

Lab 1: Identity and security

Student:

Instructor: Dr. Bin Hu

Date:

Description

This section describes the objective of the lab based on your understanding from the actual laboratory work. You should provide your **own statements**, but not copy the information listed in the "Description" section of the Lab Procedure document.

In addition, you should provide in the section the **system configuration information** including at least: Computer and its operating system version number, location (building on campus or township if off-campus) where the lab performed, and type of network connection and Internet Service Provider (ISP) used to connect to the Internet.

If any of these conditions are changed while conducting the lab procedure, you should note immediately the change in the **"Procedure and Notes"** section next to the procedure that is performed.

Procedure and Notes

1. Outcomes in a Laboratory Report should be reproducible based on the same settings. Hence, this section is an itemized list of actions and notes you took by following the lab procedure.
2. It should provide any data/information that you are asked to note or observe at that particular procedure. Some screen shots may be attached as a proof of your note and provide your instructor to determine the root cause, if your note would be incorrect.
3. If the laboratory requires to develop a program code, you should list your program code in this section.
4. You can list any problems you encountered during the lab. Discuss why it occurred and how to get it right for the correct response, if you did.

Results and Reports

1. This section is for you to discuss your final results and/or report from your laboratory experience and conclusion.
2. If there are any files prepared as the results of the lab, make sure they are accessible electronically.
3. If you use any data to test your program code, discuss the output produced by your code for the test data.

Lab 1.1: Windows Client IP Configuration (ipconfig)

- **Which options do you think would be most useful for an administrator?**
    - /all (displays full configuration details including physical MAC addresses, DNS servers, and DHCP state) and /renew or /release (to manage dynamic IP lease allocations) are the most useful.
- **Which option would you use to obtain an IP configuration from a Dynamic Host Configuration Protocol (DHCP) server?**
    - ipconfig /renew
- **What is your IP address?**
    - 192.168.1.4 (Wi-Fi adapter)
- **What is your subnet mask? What is the purpose of an IP address mask?**
    - Subnet Mask: 255.255.255.0
    - **Purpose:** The subnet mask identifies which portion of the IP address represents the network ID and which portion represents the host ID, allowing devices to determine whether communication is local (on the same LAN) or requires a router/gateway.
- **What is the MAC address of your computer?**
    - For the VirtualBox Host-Only Ethernet adapter: 0A-00-27-00-00-08. (For the Wi-Fi adapter, the physical address can be referenced via adapter details if needed).
- **What is your DNS server address?**
    - Handled via the local router/gateway configuration (typically indexed through network adapter properties or local DNS forwarders).

Lab 1.2: IP Network Configuration

Lab A: ping and tracert

- **What is the time value observed for all four replies? (pinging gateway 192.168.1.1)**
    - Round trip times were 1ms, 2ms, 2ms, and 3ms (average of 2ms).
- **What is the TTL (Time To Live) value? What does this number refer to?**
    - TTL Value: 64

- **Reference:** It defines the maximum number of router hops a packet can traverse before being discarded, preventing infinite routing loops.

- **How can you be sure that this response is actually coming from the correct computer?**
  - By pinging the exact local gateway IP (192.168.1.1) or verifying the matching MAC address resolved through the ARP cache table.

- **tracert www.kean.edu observations:**
  - **How many actual intermediate hops?** Around 6 responding active hops before reaching external transit nodes and timeouts (web.kean.edu resolves to 131.125.16.38).
  - **Which intermediate node takes the longest time?** Hop 2 (10.240.170.53) and initial ISP transit edges fluctuated up to 22ms–35ms.

- **tracert www.google.com observations:**
  - **How many actual intermediate hops?** 8 hops to reach Google's destination server (142.251.151.119).
  - **Which intermediate node takes the longest time?** Hop 3 (ubr103-ge3-0-1.cmts.nwrknj.cv.net) showed initial spikes around 35ms.

Lab B: ARP (Address Resolution Protocol)

- **At the command prompt type arp /? and press ENTER. Which option displays the current ARP entries?**
  - The -a option (arp -a) displays the current ARP cache entries.

- **Observations from arp -a:**
  - Successfully populated dynamic entries mapping local IP addresses (such as gateway 192.168.1.1) to their respective physical MAC addresses (e.g., 2c-de-32-e2-86-ff).

Reference and Acknowledgement

With or without encountering any difficulty, if you surfed the Internet to find some hints or to learn new information and/or technology to help you conducting the lab, you should include the information sources, such as the URLs, citations, or books.

If you received assistance from anyone (e.g. working together with some friend for the lab), you should acknowledge people’s contribution.